

Problem Set 10
Due: May. 10, 2021

1. Section 4.3 Problem 23.

Pat and Nat are dating, and all of their dates are scheduled to start at 9 p.m. Nat always arrives promptly at 9 p.m. Pat is highly disorganized and arrives at a time that is uniformly distributed between 8 p.m. and 10 p.m. Let X be the time in hours between 8 p.m. and the time when Pat arrives. If Pat arrives before 9 p.m., their date will last exactly 3 hours. If Pat arrives after 9 p.m., their date will last for a time that is uniformly distributed between 0 and $3 - X$ hours. The date starts at the time they meet. Nat gets irritated when Pat is late and will end the relationship after the second date on which Pat is late by more than 45 minutes. All dates are independent of any other dates.

- (a). What is the expected number of hours Nat waits for Pat to arrive?
- (b). What is the expected duration of any particular date?
- (c). What is the expected number of dates they will have before breaking up?

Solution:

- (a). Let W be the number of hours that Nat waits. By total expectation theorem, we have

$$\mathbf{E}[W] = \mathbf{P}(0 \leq X \leq 1)\mathbf{E}[W \mid 0 \leq X \leq 1] + \mathbf{P}(X > 1)\mathbf{E}[W \mid X > 1]$$

Since $W > 0$ only if $X > 1$, and given $X > 1$, W follows a uniform distribution between 0 and 1, we have

$$\mathbf{E}[W] = \mathbf{P}(X > 1)\mathbf{E}[W \mid X > 1] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

- (b). Let D be the duration of a date. We have $\mathbf{E}[D \mid 0 \leq X \leq 1] = 3$. Furthermore, when $X > 1$, the conditional expectation of D given X is $(3 - X)/2$. Hence, using the law of iterated expectations,

$$\begin{aligned}\mathbf{E}[D \mid X > 1] &= \mathbf{E}[\mathbf{E}[D \mid X, X > 1] \mid X > 1] \\ &= \mathbf{E}\left[\frac{3 - X}{2} \mid X > 1\right] \\ &= \frac{3}{2} - \frac{1}{2}\mathbf{E}[X \mid X > 1] \\ &= \frac{3}{2} - \frac{1}{2} \cdot \frac{3}{2} \\ &= \frac{3}{4}\end{aligned}$$

Therefore, by total expectation theorem,

$$\begin{aligned}\mathbf{E}[D] &= \mathbf{P}(0 \leq X \leq 1)\mathbf{E}[D \mid 0 \leq X \leq 1] + \mathbf{P}(X > 1)\mathbf{E}[D \mid X > 1] \\ &= \frac{1}{2} \cdot 3 + \frac{1}{2} \cdot \frac{3}{4} \\ &= \frac{15}{8}\end{aligned}$$

- (c). The probability that Pat will be late by more than 45 minutes is $1/8$. The number of dates before breaking up is the sum of two geometrically distributed random variables with parameter $1/8$, and its expected value is $2 \cdot 8 = 16$.

2. Section 4.4 Problem 37.

A pizza parlor serves n different types of pizza, and is visited by a number K of customers in a given period of time, where K is a nonnegative integer random variable with a known associated transform $M_K(s) = \mathbf{E}[e^{sK}]$. Each customer orders a single pizza, with all types of pizza being equally likely, independent of the number of other customers and the types of pizza they order. Give a formula, in terms $M_K(\cdot)$, for the expected number of different types of pizzas ordered.

Solution: Let X be the number of different types of pizza ordered. Let X_i be the random variable defined by

$$X_i = \begin{cases} 1 & \text{if a type } i \text{ pizza is ordered by at least one customer} \\ 0 & \text{otherwise} \end{cases}$$

We have $X = X_1 + \cdots + X_n$, and by the law of iterated expectations,

$$\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X \mid K]] = \mathbf{E}[\mathbf{E}[X_1 + \cdots + X_n \mid K]] = n\mathbf{E}[\mathbf{E}[X_1 \mid K]]$$

Furthermore, since the probability that a customer does not order a pizza of type 1 is $(n-1)/n$, we have

$$\mathbf{E}[X_1 \mid K = k] = 1 - \left(\frac{n-1}{n}\right)^k$$

so that

$$\mathbf{E}[X_1 \mid K] = 1 - \left(\frac{n-1}{n}\right)^K$$

Thus, denoting

$$p = \frac{n-1}{n}$$

we have

$$\mathbf{E}[X] = n\mathbf{E}[1 - p^K] = n - n\mathbf{E}[p^K] = n - n\mathbf{E}[e^{K \log p}] = n - nM_K(\log p)$$

3. Show how to compute $\text{cov}(X, Y)$ from the joint moment generating function of X and Y , i.e., $M_{X,Y}(s, t) = \mathbf{E}[e^{sX+tY}]$.

Solution:

$$\begin{aligned}
 M_{X,Y}(s, t) &= \mathbf{E}[e^{sX+tY}] \\
 \left. \frac{\partial^2}{\partial s \partial t} M_{X,Y}(s, t) \right|_{s=0, t=0} &= \mathbf{E}[XY e^{sX+tY}]|_{s=0, t=0} = \mathbf{E}[XY] \\
 \left. \frac{\partial}{\partial s} M_{X,Y}(s, t) \right|_{s=0, t=0} &= \mathbf{E}[X e^{sX+tY}]|_{s=0, t=0} = \mathbf{E}[X] \\
 \left. \frac{\partial}{\partial t} M_{X,Y}(s, t) \right|_{s=0, t=0} &= \mathbf{E}[Y e^{sX+tY}]|_{s=0, t=0} = \mathbf{E}[Y]
 \end{aligned}$$

$$\begin{aligned}
 \text{cov}(X, Y) &= \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y] \\
 &= \left. \frac{\partial^2}{\partial s \partial t} M_{X,Y}(s, t) \right|_{s=0, t=0} - \left(\left. \frac{\partial}{\partial s} M_{X,Y}(s, t) \right|_{s=0, t=0} \right) \left(\left. \frac{\partial}{\partial t} M_{X,Y}(s, t) \right|_{s=0, t=0} \right)
 \end{aligned}$$

4. An urn contains a white and b black balls. After a ball is drawn, it is returned to the urn if it is white; but if it is black, it is replaced by a white ball from another urn. Let M_n denote the expected number of white balls in the urn after the foregoing operation has been repeated n times.

- (a). Derive the recursive equation

$$M_{n+1} = \left(1 - \frac{1}{a+b}\right) M_n + 1$$

- (b). Use part (a) to prove that

$$M_n = a + b - b \left(1 - \frac{1}{a+b}\right)^n$$

- (c). What is the probability that the $(n+1)$ th ball drawn is white in terms of M_n ?

Solution:

- (a). Let X_n denote the number of white balls after the n th drawing. Let W_n denote the event that the n th ball drawn is white. Considering X_{n+1} given X_n , by total expectation theorem,

$$\begin{aligned}
 \mathbf{E}[X_{n+1} | X_n] &= \mathbf{P}(W_{n+1} | X_n) \mathbf{E}[X_{n+1} | X_n, W_{n+1}] + \mathbf{P}(W_{n+1}^c | X_n) \mathbf{E}[X_{n+1} | X_n, W_{n+1}^c] \\
 &= \frac{X_n}{a+b} \cdot X_n + \left(1 - \frac{X_n}{a+b}\right) \cdot (X_n + 1) \\
 &= \left(1 - \frac{1}{a+b}\right) X_n + 1
 \end{aligned}$$

Using the law of iterated expectations, we have

$$\begin{aligned} M_{n+1} &= \mathbf{E}[X_{n+1}] \\ &= \mathbf{E}[\mathbf{E}[X_{n+1} \mid X_n]] \\ &= \mathbf{E}\left[\left(1 - \frac{1}{a+b}\right)X_n + 1\right] \\ &= \left(1 - \frac{1}{a+b}\right)\mathbf{E}[X_n] + 1 \\ &= \left(1 - \frac{1}{a+b}\right)M_n + 1 \end{aligned}$$

(b). Note that we have the boundary condition $M_0 = a$ for the recursive equation in (a).

$$\begin{aligned} M_n - (a+b) &= \left(1 - \frac{1}{a+b}\right)M_{n-1} + 1 - (a+b) \\ &= \left(1 - \frac{1}{a+b}\right)(M_{n-1} - (a+b)) \\ &= \left(1 - \frac{1}{a+b}\right)^n (M_0 - (a+b)) \\ &= \left(1 - \frac{1}{a+b}\right)^n (-b) \end{aligned}$$

Therefore,

$$M_n = a + b - b \left(1 - \frac{1}{a+b}\right)^n$$

(c). By total probability theorem,

$$\begin{aligned} \mathbf{P}(W_{n+1}) &= \sum_x \mathbf{P}(W_{n+1} \mid X_n = x) \mathbf{P}(X_n = x) \\ &= \mathbf{E}[\mathbf{P}(W_{n+1} \mid X_n)] \\ &= \mathbf{E}\left[\frac{X_n}{a+b}\right] \\ &= \frac{M_n}{a+b} \end{aligned}$$